SHREYAS S S

Civil Engineering Undergraduate | Co-Founder, Aptimizer | AI-Driven Planning for the Built Environment

JP Nagar, Bengaluru, India • +91 72047 78542 • yesitshreyas@gmail.com

PROFILE

Final-year Civil Engineering student and co-founder of **Aptimizer**, an AI-assisted decision support system for planning and code-compliance of multi-tower residential complexes. Combines core civil engineering knowledge with self-taught programming in **Python, C/C++, Java and SQL** and hands-on computer vision work. Brings rare range for an engineer — site exposure at an international formwork firm plus real-world brand marketing. Speaks **7 languages** and explains complex ideas simply. Targeting roles in **proptech, smart infrastructure, BIM automation and AI for construction**.

FLAGSHIP PROJECT

Aptimizer — Co-Founder & Lead Developer 2025 – Present

Development of Aptimizer: An AI-Assisted Decision Support System for Planning and Compliance Assessment of Multi-Tower Residential Complexes • Final-year project, DSCE (Guide: Dr. Geetha L)

- Co-lead a **4-member team** building a staged platform that takes a residential project from raw site to compliant, buildable layout — replacing slow manual spreadsheet-and-CAD feasibility work.
- **Delivered Stage 1 (Site Analysis / GIS):** automated plot assessment and feasibility, tested across **10 plots**.
- Engineered the compliance core: **FAR / FSI** limits, setbacks and minimum spacing, plus **Carpet, Built-Up and Super Built-Up area** calculations aligned to Indian building codes (IS 3861).
- Modelled unit allocation across 2BHK, 3BHK, 4BHK and duplex mixes with corridor, lift, staircase and parking constraints, including duplex-stacking and shared common-area logic.
- Building toward automated **BOQ generation**, AI-assisted layout design, green-building checks and **BIM / Revit** integration across Stages 2–5.

Real-Time Face & Eye Detection System — Computer Vision Academic Project

- Built a real-time pipeline in **Python and OpenCV** detecting faces and eyes from a live webcam stream with Haar Cascade classifiers, tuned for low-latency inference.

EXPERIENCE

Marketing Consultant — Chebel Patisserie & Café Jul – Aug 2025

Premium artisan patisserie & café • JP Nagar, Bengaluru

- Led **influencer outreach** for the brand: identified relevant food and lifestyle creators, cold-called them and pitched paid and barter collaborations.
- Served as the brand's point of contact for creators, following up to confirm collaborations and aligning them with the café team.

Systems & Operations Intern — Tunnels & Formworks International S.R.L. Jan 2025 • 15 days

International formwork company (Italy • India • Australia) • Bengaluru

- Studied formwork systems, concreting techniques and fabrication methods used on large-scale infrastructure projects.
- Observed structural workflows, material handling and site coordination, noting opportunities for material efficiency and waste reduction across construction cycles.

SKILLS

Programming: Python, C, C++, Java, SQL, HTML

AI & Computer Vision: OpenCV, Haar Cascade object detection, real-time video processing, machine learning fundamentals

Engineering & Design: CAD, SketchUp, GIS-based site analysis, FAR/FSI & area compliance, space planning, Adobe tools, MS Office

Languages (7): English (C2), Hindi (C2), Kannada (B2), Marathi (B2), French (B1), Spanish (A1), Korean (A1)

Marketing & Communication: Influencer outreach, brand collaborations, pitching, peer tutoring

EDUCATION

B.E. Civil Engineering — Dayananda Sagar College of Engineering 2023 – 2027

Pre-University, Science (PCMC) — Christ Junior College 2021 – 2023

ICSE — The New Cambridge High School 2021

ACHIEVEMENTS & LEADERSHIP

- **3rd Prize** — Idea-Pitching Challenge (Infrathon), DSCE — pitched Aptimizer as a startup venture.
- **1st Place** — Silencieux & Épelons! French events, Prerana Language Association, Christ Junior College.
- **1st Place** — Talent's Day Fashion Show, DSCE • **3rd Place** — XITE Group Dance • **Student Coordinator** — College Magazine 2022, Christ Junior College.
- **Community work:** 10-day survey camp, Swachh Bharat drive, traffic awareness, teaching kids, digital literacy.